

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		Substitution: $35 \times 10 \times 0.8$ (1) PE = 280 [J] (1)	1	1		2	2	
		(ii)		280 [J] ecf		1		1		
		(iii)		Substitution: $280 \text{ ecf } = \frac{1}{2} \times 35 \times v^2$ (1) Rearrangement: $v^2 = \frac{2 \times 280}{35}$ (1) $v = 4$ [m/s] (1) Answer of 16 only award 2 marks Alternative: Substitution into: $v^2 = u^2 + 2ax$ i.e. $v^2 = 0$ (1) + $2 \times 10 \times 0.8$ (1) $v = 4$ [m/s] (1)	1	1 1		3	3	
	(b)	(i)		Selection of $x = ut + \frac{1}{2}at^2$ (1) $x = [0 +] \frac{1}{2} \times 10 \times 0.5^2$ (1) $x = 1.25$ [m] (1)	1 1	1		3	3	
		(ii)		Substitution and rearrangement: $\frac{6.4}{3.0}$ (1) $F = 2.1$ [N] (1)		2		2	2	
				Question 4 total	4	7	0	11	10	0